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NBIOT METERS PROTOCOL

Abbreviation

- R - read
- W - write
- E - execute
- S - Single
- O - Optional
- Bn - base name Object code, e.g. /3/0

Protocol structure

Name	Length	Sample data (hex)	Remark
Version number	2 bytes	01 01	V1.01
Message type	1 byte	00	The message needs confirmation
Message code	1 byte	02	It's used to update a kind of resource
Message ID	2 bytes	01 01	Message sequence number=0x0101
Data field format	1 byte	3C	3C:it indicates that the data field is in CBOR format
Data field length	2 bytes	00 02	The length of the data field
Delimiter	1 byte	FF	
Data field	Nb/e	AA 55	Terminal data
Check code	2 byte		Using the data above, calculate the CRC-16 checksum per the CGITT algorithm (Check appendix).

Details description

1. All data items are transmitted with the high byte first and the low byte after
2. Downlink data packaging on the server side, except for the message serial number, data field length, data field data, and check code, which need to be modified; other data items can be filled in according to the sample data.
3. On the server side, the message sequence number of the downlink response must be consistent with the corresponding uplink message
4. The length of the data field should be consistent with the length of the data field filled in
5. If there is no user data to be sent in the downlink response, the length of the data field can be 0, and a null data frame can be sent as a response.

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Overview

- The protocol is based on JSON, and the listed resource numbers can be selected according to requirements
- Each device object contains different content. Therefore, you need to identify the device's object before parsing it.

Headers

Here is the header information before the data field in HEX format, with a fixed-length field

Hexadecimal data: 01 01 00 02 E0 8B 3C

```
"bver":1.01,              //0 bver/versions 2 bytes
"msgType":0,              //1 msgType 1 byte
"code":2,                 //Function code/response code 1 byte
"messageId":57483         //3 messageId 2 bytes
"payloadFormat":60,       //4 message format 1 byte
```

·0 Version hex

Fixed as 01 01

·1 Message type hex

0-The message needs confirmation
1-The message does not require confirmation
2-Response message
3- Reset message

·2 Function code/response code

The device proactively reports to the platform with the function code 0x02.

The device responds to the platform-setting instruction, with function code 0x44.
The platform issues a setting instruction, with function code 0x45.

The platform delivers calibration; the function code is 0x03.

·3 Message ID (random number)

The name of the message delivered by the platform must match the ID of the message proactively reported by the device.

The device's message ID must match the platform's message ID for the setting command.

·4 Message format

60 - CBOR: it's a concise binary object representation

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Object definition

Here is the object in the data field, convert CBOR to JSON format

Object number: bn: /3/0

It provides a series of device-related information to inform the server

```
[{
{"Manufacturer”: Ubuntu/Hac},          // 0    Manufacturer name

{"Model": H5lA},                       // 1    Model

{"Serial Number":1234567890},          // 2    Serial number
{"Firmware Version": null},            // 3    Firmware version no.
{"reboot":0},                          // 4    reboot
{"Factory Reset": null},               // 5    Factory Reset
{"Available Power Sources":1}, // 6  Power supply type
{"Power Source Voltage":360},          // 7    Power supply Voltage
{"Power Source Current”:null}, // 8          Power supply Current
{"Battery Level":95},                  // 9    Battery Level
{"Memory Free": null},                 // 10 Memory Free (”Error
Code":1},                              // 11   Error Code
("Reset Error Code":0},                // 12 Reset Error Code
("Current Time":1527125907},           // 13   Current Time Unix Time
{"UTC Offset":UTC+8},                  // 14  Time zone [ISO 8601
{"Timezone":+0800},                    // 15  Time zone IANA
{"Binding Modes":U},                   // 16 Binding Modes
{"Device Type":1},                     // 17   Device Type
{"Hardware Version":PV1.0},            // 18   Hardware Version
{"Software Version":Vl.01},            // 19   Software Version
{"Battery Status":0},                  // 20   Battery Status
```

{"Memory Total":null}, // 21 Memory Total

{"ExtDevInfo":null}, // 22 Extend device information/connection URI /xx/x/xx

{"Message Sequence":1}, // 23 Message Sequence

{"Working hours":100}, // 24 Meter Working hours

{"query information": 1}, // 40 query information

}]

·0 Manufacturer name Ubuntu/Hac

·1 Module name NBh

·2 R Multiple Optional String

·3 Firmware version no.

·4 Reboot

Reset the MCU

·5 Factory reset (Not supported in this firmware)

·6 Power supply type R Multiple Optional Integer 1-7

1 - Internal battery (it's not currently supported)

2 - External battery can be replaced

4 - Power supply by Ethernet (it's not currently supported)

5 - USB (it's not currently supported)

6 - AC power supply (it's not currently supported)

7 - Power supply by solar energy (it's not currently supported)

·7 Power supply voltage R Multiple Optional Integer l0mV

·8 Power supply current R Multiple Optional Integer mA (it's not currently supported)

·9 Battery level R Multiple Optional Integer 0-100 % (it's not currently supported)

·10 MCU memory remaining (it's not currently supported)

·11 Error code

1 - no error

·12 Reset error message code (it's not supported temporarily)

·13 Current time R Single Optional Time

Unix timestamp, or Unix time, POSIX time, is a way of time representation, it's defined as 00:00:00 on January 01, 1970, Greenwich Mean Time, the total number of seconds since the present. Unix timestamps are not only used in Unix systems and Unix-like systems, but also widely used in many other operating systems

Example: 1527125907 (2018/5/24 9:38:27)

·14 UTC time zone RW Multiple Optional String UTC+X

·15 Time zone IANA (it's not supported temporarily)

·16 Binding mode (it's not supported temporarily)

·17 Device type R Single Optional Integer 1
– prepaid water meter

·18 Hardware version no.

·19 Software version no.

·20 Battery status R Single Optional Integer 0-6 (Currently 0,4 are valid This value is only valid for the external battery of the device (the power type is 1)

0 – battery is normal
1 - Battery is charging now
2 – The charging is complete, the battery is fully charged, and is still charging
3 – The battery is damaged
4 – Low battery
5 – The battery is not installed
6 – The battery type is unknown; the battery information is not available.

·21 Full memory size (it's not currently supported)

·22 Extended device information

·23 Message packet serial number R Single Optional Integer

The value ranges from 0 to 256. The device accumulates only when reporting regularly; it does not accumulate during retransmission.

·24 Meter working hours R Single Optional Integer
Unit: hour

·40 Information query (it's not supported temporarily) 1
- query information

Object number: bn:/70/0

The content is added to the protocol queue for cached execution.

```
[{
{"mid":"65536"},Value 0-65536 2 bytes
}]
```

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Object number: bn:/80/0

Meter Basic: Meter basic information

```
[{
// 0 meter type
        {"meter type":"1"},
// 1 measurement model
        {"Measurement model":0},
// 2 The pulse constant
        {"PN":0},
// 3 Meter size
        {"DN":20},
// 4 Commonly used traffic(Q3)
        {"Q3":550},
// 5 Q3/Q1=1000：Common flow ratio minimum flow =1000, that is, minimum flow Q1=Q3/1000=0.025m³/h
        {"Q3/Q1":400},
// 6 Measurement error state
        {"Measurement fault status":0},
// 7 Maximum meter reading
        {"Max Reading":99999999},
// 8 instantaneous flow rate
        {"Real flow rate":99999999},
// 9 instantaneous power
        {"Real Power":99},
// 10 cumulative discharge temperature
        {"Accumulated heat":99999},
// 11 water inlet temperature
        {"Inlet water temperature":80},
// 12 return water temperature
        {"Back water temperature":60},
// 13 Forward accumulated flow
        {"Positive Flow":99999999},
// 14 Reverse cumulative flow
        {"Reverse Flow":99999999},
// 15 Peak flow rate
        {"peakFlowRate":1200},
```

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```
// 16 meter reading (cumulative discharge)

        {"meter Reading":999999999},


// 17 One day dense data

        {"Dense
        data":h'006A13070A131110000000011388138813881388138813881388138813881388138813881388138813881388138813881388138813881
        3881388138813881388138813881388138813881388138813881388138813881388138813881388138813881388138813881388138813
        881388'},

// 18 Frozen data frame

        {"Frozen data":
        h'00D21306030000000013060300000000130603000000001306030000000013060300000000130603000000001306030000000013060
        3000000001306030000000013060300000000130603000000001306030000000013060300000000130603000000001306030000000013
        0603000000001306030000000013060300000000130603000000001306030000000013060300000000130603000000001306030000000
        013060300000000130603000000001306030000000013060300000000130603000000001306030000000013060300000000'},


// 19 Data is frozen when it was read

        {"Read day Frozen data":0]},

// 21 Meter reading time

        {"Reading Meter Time":1517983759},

// 23 Available water allowance

        {"Available Water Allowance":999},

// 24 Available water allowance alarm value

        {"Available Water Allowance Alarm value":999},

// 26 Overdraft amount

        {"Overdraft volume":999},

// 27 Daily intensive data acquisition cycle(min)

        {"Dense data Cycle":30},

// 28 Read daily frozen data for a certain day

        {"Read Dense data": 190719},

// 29 Read monthly frozen data

        {"Read Month Frozen data": 0},

// 30 Read a day's intensive data

        {"Read Dense Frozen data": 0},

// 31 Enable the prepaid function

        {"payment function": 0},

// 32 Enable intensive traffic sampling

        {"Dense Frozen function": 0},

// 33 Thermometer temperature difference

        {"temperature difference": 20},
```

```
// 34 pressure value
        {"pressure value": 20} ,
// 35 temperature value
        {"temperature value": 2532},
// 36 Table end status of the electromagnetic water meter
        {"meter status": 0},
// 37 battery voltage
        {"meter battery voltage": 368},
// 38 The cumulative amount of cold
        {"Accumulated cold water": 1000},
// 40 Querying information on the table
        {"query meter information": 1},
// 41 Latitude
        {"latitude information": "N22.3468202"},
// 42 Longitude
        {"longitude information": "E113.5608126"},
// 43 altitude
        {"altitude": 60},
}]
```

·0 Meter type R Multiple Optional String

·1 Metering mode R Single Optional Integer

0 - Dual reed switch
1 - Single reed switch
2 - Dual hall
3 - Direct reading meter
4 - Non-magnetic inductive
5 - Non-magnetic coil
6 - Three hall
7 - Single hall

·2 PN Pulse constant R Single Optional Integer

0 - Direct meter reading
1 - 1L/P
2 - l0L/P
3 - l00L/P
4 - l000L/P
5 - 0.5L/P

6 - 5L/P

·3 Meter DN size

·4 Common flow

·5 Minimum flow

·6 Meter status word R Single Optional Integer

LSB

bit 0: Historical magnetic attack

bit l: Measurement data error

bit 2: magnetic attack

·7 The maximum reading WR Single Optional Integer

·8 Instantaneous flow R Single Optional Integer Unit: L/h

·9 Instantaneous power R Single Optional Integer Unit: W

·10 Accumulated heat R Single Optional Integer Unit: Wh

·11 Inlet water temperature R Single Optional Integer Unit: 0.01°C

·12 Return water temperature R Single Optional Integer Unit: 0.01°C

·13 Positive accumulated flow R Single Optional Integer

·14 Negative accumulated flow R Single Optional Integer

·15 The highest peak flow rate (it's not currently supported)

·16 Meter readings R Single Optional Integer 0-the maximum reading L

·17 Daily dense data R Optional Bytestring

Record daily dense data, which is the flow difference within a single collection period (take 30 minutes as an example). The first two bytes indicate the length of the data field, **0066,** which means 102 bytes, and the next 6 bytes are the start time of the collection.YYMMDDHHMMSS, the next 4 bytes represent the start flow pulse number and every two bytes after the start time is the difference of the flow pulse number collected after the start time (signed), 1388 means the flow rate is 5000

Note: If a value of 2 PN is reported, the real flow is the number of pulses of the dense flow data multiplied by the PN value when analysing the platform. If the PN value is not reported, the number of pulses of the dense flow is multiplied by 1; for example, the PN value is 10. The number of dense flow pulses is 5000, and the dense data is 50000L

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Name	length	Sample data(hex)	remark
Data field length	2 bytes	00 6A	106 bytes
Starting time	6 bytes	13 07 0A 13 11 10	Start time: 19:17:16 on July 10, 2019
Number of initial flow pulses	4 bytes	00 00 00 001	Indicates that the start traffic is 1L
Flow pulse number difference	2*48 byte	13 88	(signed) indicates that the difference in the number of flow pulses is 0x1388 = 5000 (L)

·18 Freeze data response (daily freeze, monthly freeze) R Optional Bytestring

Record frozen data, which is a byte stream object, each is fixed 7 bytes, including (year, month, day, hexadecimal) 3 bytes plus 4 bytes of flow data, the length is not limited, the first two bytes are the data length, for example: 00 D2 means there are 210 bytes of data behind

Name	Length	Sample data(hex)	Remark
Data field length	2 bytes	00 D2	It represents 210 bytes
Date flow	7*n bytes	13 06 03 00 00 13	It means that the frozen date is June 3, 2019, and the frozen flow is 0x00001388 =5000 (L)

·19 Read daily frozen data R Single Optional Integer

0 - read recently frozen data

1~8 read the current month, the previous one month, the previous 2 months...the previous 8 months and the daily frozen data

·21 Meter reading time R Single Optional Integer

·22 Water meter no.

·23 Available water margin RW Optional Integer (with prepaid function)

Unit:L

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·24 Available water margin alarm value RW Optional Integer (with prepaid function)

Unit:L

·26 Overdraft amount Optional Integer Unit: L (with prepaid function)

The amount of overdraft water that can continue to be used after the remaining available water is used up. If it's more than 0, it means overdraft is possible; if it's equal to 0, it means no overdraft; the valve will be closed immediately after using it up; and 999999999 means no closing.

·27 Daily dense data collection cycle RW Optional Integer Unit: min

Unit: min; the minimum is 15, it's 15 by default, it must be a divisor or multiple of 60, the daily dense data needs to be collected according to this cycle; it can be set to: 15, 20, 30, 60, 120, 240, 360, 480, 720

·28 Read a certain frozen data time (daily /month1y frozen) R Single Optional Integer

It is an integer variable, such as 240718 (decimal), which means read the daily frozen data on July 18, 2024, including monthly frozen data;2407 (decimal), which means to read the monthly frozen data of July 2024.

·29 Read monthly frozen data R Single Optional Integer

0~20 Read the current year, the previous l year, the previous 2 years... the previous 20 years, of the monthly frozen data

·30 Read dense frozen data R Single Optional Integer

It's expressed as an integer variable, such as 240718 (decimal), which means read the dense frozen data on July 18, 2024.

·31 Payment function enable bit: R Single Optional Integer (with prepaid function) 0 - disable

l - enable

·32 Daily dense flow sampling enable bit R Single Optional Integer 0 - disable

l - enable

·33 Heat meter temperature difference R Single Optional Integer（if temp sensor is available)

Unit: 0.0 1 °C

·34 Pressure value R Single Optional Integer (if pressure sensor is available)

Unit: KPa

·35 Temperature value (if temp sensor is available)

Unit: 0.0 1 °C

·36 Status of electromagnetic water meter R Single Optional Integer

0 - OK

l - Alarm / Tamper/Critical

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·37 Battery voltage R Single Optional Integer

Unit: 0.01V

·38 The cumulative amount of cold R Single Optional Integer Unit: Wh(With temp sensor)

·40 Meter information query

1 - Query meter information

·41 Latitude information R Multiple Optional String

The value is in the format of direction + degree + minute, and degrees and minutes are separated by periods (.). Direction: N-north, S-South

Description: Using WGS84 coordinate system earth coordinate system, international general coordinates DD.mm +(fractional decimal)

·42 Longitude information R Multiple Optional String

The format is as follows: direction + degrees + minutes, and degrees and minutes are separated by periods (.). Direction: E- east, W-west

Description: Using WGS84 coordinate system earth coordinate system, international general coordinates DD.mm +(fractional decimal)

·43 Altitude R Single Optional Integer

Unit: m

Object number：bn:/81/0

```
[{
  {"Valve Control":0},                // 0 Valve Control
  {"Valve Current status":0},         // 1 Valve status
  {"Valve fault status":0}            // 2 Valve failure condition
  {"Valve type":0}                    // 3 valve type
  {"Valve overturn Period":1}         // 4 Regularly dredge valve cycle
  {"ATK Close Valve":1}               // 5 Magnetic attack valve off enable position
  {"Force Valve Control":0},          // 6 Forced control valve
  {"Valve Control Timeout":0},        // 7 Set the valve timeout period
  {"Valve opening angle":90},         // 8 Set valve opening Angle
  {"query Valve information":1}       // 40 information and telephone directory
}]
```

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·0 Valve control word RE Single Optional Integer

0 - valve open

1 - valve close

·1 Valve status R Single Optional

Integer 0 - valve open status

1 - valve close status

·2 Valve failure status R Optional Integer

0 - normal

1 - failure

·3 Valve type RW Optional Integer

0 - two-wire

1 - five wire-close in place

2 - no valve

3 - five wire-disconnected in place

·4 Regularly dredge the valve cycle RW Optional

Integer Unit: times/month

Cycle by default: 1 time (the setting range:0-4)

It's set to 0, which means the valve-dredging function is disabled.

·5 Magnetic attack valve close enable bit RW Optional

Integer 0 - disable

1 -enable

·6 Forced control valve RE Single Optional Integer

0 - Forcibly open the valve

1 - Forcibly close the valve

2 - Cancel the mandatory control

·7 Set the valve timeout period RW Optional

Integer Unit: second

Setting range: < 4000s

·8 Set valve opening Angle RW Optional

Integer Units: ℃

Setting range: 0~90

·40 Valve information query (it's not supported currently)

1 - Query information

Object number：bn:/82/0

Abnormal alarm: Abnormal alarm

```
[{

 {"Magnetic attack status":0},       // 0
 {"happened Magnetic attack":0}      // 1
 {"Anti demolition":0}               // 2
 {"happened Anti demolition":0}      // 3
 {"leak status":0}                   // 4
 {"over flow status":0}              // 5
 {"stop status":0}                   // 6
 {"Water error code":0}              // 7
 {"heat error code":0}               // 8
 {"Ultrasonic error code":0}         // 9

 {"Available Water Alarm":0},        // 11
}]
```

·0 Magnetic attack status R Single Optional Integer

0 - none

1 - yes

·1 Happened Magnetic attack R Single Optional Integer

0 - none

1 - yes

·2 Anti demolition R Single Optional Integer

0 - none

1 - yes

·3 Happened Anti demolition R Single Optional Integer

0 - none

1 - yes

·4 Leak status R Single Optional Integer

0 - none

1 - yes

·5 Over flow status R Single Optional Integer

0 - none

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1 - yes

·6 Stop status R Single Optional Integer

0 - none

1 - yes

·7 Water error code R Single Optional Integer

·8 Heat error code R Single Optional Integer

The 16-bit fault code is customised. The low byte is ST1 and the high byte is ST2, for example:

·ST1

Bit0-bit1: undefined;

BIT2: battery undervoltage 1, normal 0; Bit3-bit7: undefined;

·ST2

BIT0: battery undervoltage 1, normal 0;

BIT1: Water inlet temperature sensor alarm 1, normal 0; BIT2: backwater temperature sensor alarm 1, normal 0; BIT3: flow sensor alarm 1, normal 0;

Bit4-bit7: undefined;

·9 Ultrasonic error code R Single Optional Integer
User-defined description:

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
Blue temperature	Red temperature	Air tube/transducer	EE sensor	keep	Battery voltage	keep	keep
0 Normal 1 Abnormal	0 Normal 1 Abnormal	0 Normal 1 Abnormal	0 Normal 1Abnormal		0 Normal 1Abnormal		

·10 Reverse Flow status R Single Optional Integer

0 - none

1 - yes

·11 Available Water Alarm (Prepaid function)

Assume that after a user recharges, the "water allowance" is 20 units, and the "water allowance alarm value" is set to 5 units.

When the meter "available water allowance" reaches 5 tons, the alarm condition is triggered.

0 - Sufficient margin

1 - Insufficient allowance

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Object number：bn:/84/0

·0 Report cycle WR Single Optional Integer

```
[{

  {"Reporting frequency":86400},      // 0

 {"Delivery frequency":86400},
  {"randomized Delivery Window":130}, // 1

  {"retries":3},                       // 2

  {"retryPeriod":1200}                 // 3

  {"Start reporting date":10150415}      //
  4
  {"Reporting interval":2400}          // 5

  {"RF work mode":0}                  // 10

  {"RF channel":0}                    // 11

}]
```

It's 86400s by default, the range can be set: 0 or 900~ 2592000 (30 days) , 0 is storage mode

Unit: second

·1 Report time slot (the modification is not supported temporarily)

It's 130s by default.

·2 Re-transmission times

The range can be set: 0-2

·3 Re-transmission cycle

It's 1200s by default

·4 The starting meter reading time and closing meter reading timeWR Single Optional Integer (it's supported by temp sensor)

It's expressed as an integer variable, such as 101504 15 (decimal), which means that the starting time of meter reading is October 15 and the closing time of meter reading is April 15

·5 Report interval, discrete start time and end time

It's expressed as an integer variable: the high two digits indicate the starting time, and the low two digits, multiplied by 30 minutes, indicate a discrete time period. For example, 2004 (decimal) indicates that the start time is 20 o'clock, the duration is 2 hours, and the end time is 22 o'clock.

The range can be set: 0:00:00 23:00:00

·10 RF working mode

0 - report once

1 - continuously reporting

·11 RF working mode

Set the RF working channel group of the meter, which can be set to 1, 2, 3

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Object number : bn:/99/0

NB Delivery: NB interactive information

```
[{

 {"NB Model Ver":"BC95B5JBR01A02W"},  // 0 NB module version number
 {"imei":863703030450097},            // 1
 {"imsi":460111174804625},            // 2
 {"iccid":89860317492039342173},      // 3
 {"IP":"111.111.111.111:5683"},       // 4  IP address    the port number
 {"APN":"ctnb"},                      // 5 APN
 {"plmnID":46011},                    // 6 plmnID
 {"BandIndicator":5},                 // 7 frequency band
 {"EARFCN":2506},                     // 8 frequency point, the value range is :0~65535
 {"CellID":125124433},                // 9 the physical community ID
 {"pci":504},                         // 10 community ID
 {"rsrp":-180},                       // 11 rsrp LTE refer to signal received power
 {"rsrq":-30},                        // 12 rsrq LTE refer to signal received quality
 {"rssi":-129},                       // 13  rssi    the strength of receiving signal
 {"snr":-10},                         // 14 SNR (Signal to Noise Ratio)
 {"ECL":2},                           // 15 Transmit coupling coverage level
 {"TX power":10},                     // 16 TX power
 {"TX time":12856},                   // 17 TX time
 {"RX time":12856},                   // 18 RX time
 {"CSQ":19}                           // 19 Module single quality
 {"Read NB information":1}            // 20 query NB module information
 {"PSM":0}                            // 21 PSM
 {"Network Protocol":0}               // 23 NWP
}]
```

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·0 Module version no R Multiple Optional String

·1 Module IMEI R Multiple Optional String

·2 Card information IMSI R Multiple Optional String

·3 Card information ICCID R Multiple Optional String

·4 IP address port number RW Multiple Optional String

·5 APN RW Multiple Optional String

·6 PLMN R Multiple Optional String

·7 Band R Multiple Optional String

·9 CellID R Multiple Optional String

·11 RSRP R Multiple Optional String

·13 RSSI R Multiple Optional String

·14 SNR R Multiple Optional String

·19 CSQ R Multiple Optional String

·20 Query NB information R Single Optional Integer
1 - Query information

·21 PSM enable bit RW Single Optional Integer

- 0 - Disable
- l - Enable

·23 Network protocol R Single Optional Integer (it's not currently supported) 0
- COAP

1 - UDP

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Appendix

CRC16/AUG-CGITT Reference code :

JAVA language code

```
 {

public static int crc_16_CCITT_False(byte[] bytes, int length)

        {int crc = 0x1D0F; // initial value

        int polynomial = 0x1021; // poly

        value

        for (int index = 0; index < length; index++)

            {byte b = bytes[index];

            for (int i = 0; i < 8; i++) {

                boolean bit = ((b >> (7 - i) & 1) == 1);

                boolean c15 = ((crc >> 15 & 1) == 1);

                crc <<= 1;

                if (c15 ^ bit)

                    crc ^= polynomial;

            }

        }

        crc &= 0xffff;

        //Prints the hexadecimal value of String

       String strCrc = Integer.toHexString(crc).toUpperCase();

        System.out.println(strCrc);
```